An Integrated Explainable AI Platform for Academic Guidance, Careers, and Wellbeing

R26-IT-073

Project Proposal Report

Rusiru L.W.S –IT22220288

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Rusiru L.W.S	IT22220288	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Ms. Thilini Jayalath		

ABSTRACT

Every year, over 340,000 Sri Lankan students face one of the most critical decisions of their lives after the GCE Advanced Level examination: choosing the right university degree. Traditional guidance systems rely heavily on grades and Z-scores, often ignoring the student's dream career, personality traits, entrepreneurial mindset, emotional wellbeing, lifestyle constraints (such as family attachment, travel tolerance, and stress tolerance), and long-term job market realities. This leads to degree–career mismatches, higher dropout rates, underemployment, and increased youth unemployment (currently ~19% for ages 15–24 in 2025).

Despite advances in AI-based career guidance, no existing system in Sri Lanka or similar contexts combines **backward career intelligence** (starting from the dream job and working backwards) with **holistic multi-parameter analysis** that includes academic performance, personality, lifestyle compatibility, district-based admission factors, and future job sustainability all delivered through Explainable AI (XAI) with counterfactual guidance and long-term roadmapping in a trilingual, Singlish-aware mobile interface.

This research proposes an **Academic Pathway and Career Intelligence Predictor**, a React Native mobile application that starts from the student's dream job, applies backward-chaining probabilistic modelling (XGBoost), integrates holistic analysis, uses SHAP and DiCE for transparent explanations and "if-then" scenarios, and generates personalized long-term academic/career roadmaps with actionable bridging plans. The system is built using FastAPI backend, Supabase database, and multilingual support for Sinhala, Tamil, and English.

Expected outcomes include ≥85% prediction accuracy, ≥80% user satisfaction (SUS), and clear, trustworthy guidance that reduces mismatched degree choices. The solution aligns with SDGs 4 (Quality Education) and 8 (Decent Work), offering significant social impact by empowering youth and commercial potential through a freemium mobile app for Sri Lanka's large A/L cohort.

Keywords: Explainable AI, Career Guidance, Backward Chaining, Multi-Parameter Analysis, Sri Lankan Education, Singlish Processing, Long-term Roadmapping

ACKNOWLEDGEMENT

I would like to express my deepest gratitude to everyone who supported me throughout the development of this project, "Academic Pathway and Career Intelligence Predictor" one of the core modules of the Integrated Explainable AI Platform for Academic Guidance, Careers, and Wellbeing.

First and foremost, I am immensely thankful to my Supervisor, Ms.Thilini Jayalath - Senior Lecturer, for her continuous guidance, valuable feedback, and encouragement. Her insightful suggestion to incorporate backward career intelligence, holistic multi-parameter analysis, and lifestyle compatibility factors significantly deepened the scope and novelty of this component. Without her direction, this work would not have reached its current level of maturity.

I am equally thankful to my Co-supervisor, Ms. Jenny Kishara - Senior Lecturer, for her constant support, technical advice, and constructive criticism. Her expertise and encouragement were instrumental in refining the system architecture, XAI integration, and mobile implementation aspects of the project.

I am also grateful to the Department of Information Technology and the university for providing the academic environment, resources, and ethical framework that made this research possible.

Special thanks go to my family and friends for their unwavering support, patience, and understanding during the many late nights and weekends spent on this project.

I also appreciate the open-source community and the developers of tools such as React Native, FastAPI, XGBoost, SHAP, DiCE, and Supabase - their excellent libraries made the implementation feasible within the time and resource constraints of a final-year project.

Finally, I thank all my classmates and batchmates who shared their thoughts, participated in informal testing, and provided constructive feedback on early prototypes.

This project is a collective effort in many ways, and I am sincerely grateful to everyone who contributed directly or indirectly to its success.

TABLE OF CONTENT

DECLARATION……2

ABSTRACT…… 3

ACKNOWLEDGEMENT…… 4

TABLE OF CONTENT ……5

LIST OF TABLES…….6

LIST OF FIGURES ……6

LIST OF APPENDICES. ……6

LIST OF ABBREVIATIONS ……7

1. INTRODUCTION ..……8

1.1 Background and Context..…..……8

1.2 Literature Review ……9

1.3 Research Gap…… 11

2. OBJECTIVE……..12

2.1 Main Objective…….12

2.2 Specific Objective……12

3. METHODOLOGY……13

4. HIGH-LEVEL SYSTEM ARCHITECTURE ..……16

5. USER REQUIREMENTS…….19

6. COMMERCIALIZATION PLAN…….21

7. BUDGET AND JUSTIFICATION……24

8. WORK BREAKDOWN STRUCTURE (WBS). ……26

9. GANTT CHART ……28

10. REFERENCES LIST. ……29

11. APPENDICES. ……34

LIST OF TABLES

Table 1:Justification Table…….14

Table 2 :Sample Persona……21

Table 3 :Sample Persona……22

Table 2 :Cost Estimation Table ……23

Table 4:Budget and Justification Table……..24

Table 5:Task Breakdown Table……..26

Table 6:Risk Analysis Table. ……34

LIST OF FIGURES

Figure 1:System Architecture Diagram…….17

Figure 2:Gantt chart ……28

Figure 3:Verification Data Flow Diagram…….33

Figure 4:Mentorship Platform Relationship Diagram ……33

Figure 5:Preliminary UI Mockups ……..34

LIST OF APPENDICES

Figure 4.1: Verification Data Flow Diagram ……33

Figure 5.2: Mentorship Platform Relationship Diagram……33

Figure 6.3: Preliminary UI Mockups ……34

LIST OF ABBREVIATIONS

Abbreviation	Full Form / Meaning
A/L	Advanced Level (GCE A/L examination in Sri Lanka)
O/L	Ordinary Level (GCE O/L examination in Sri Lanka)
AI	Artificial Intelligence
XAI	Explainable Artificial Intelligence
SHAP	SHapley Additive exPlanations (XAI technique)
DiCE	Diverse Counterfactual Explanations (counterfactual reasoning library)
XGBoost	eXtreme Gradient Boosting (machine learning model)
API	Application Programming Interface
REST	Representational State Transfer (API architecture)
UI	User Interface
UX	User Experience
SDG	Sustainable Development Goal
UGC	University Grants Commission (Sri Lanka)
Z-score	Standardized score used for university admission in Sri Lanka
NLP	Natural Language Processing
Singlish	Sinhala-English mixed language (Romanized Sinhala)
i18n	Internationalization (support for multiple languages)
JWT	JSON Web Token (authentication method)
HTTPS	HyperText Transfer Protocol Secure
DSR	Design Science Research (research methodology)
SUS	System Usability Scale (usability evaluation tool)
NPS	Net Promoter Score (satisfaction metric)
F1	F1-score (model performance metric)
BSc	Bachelor of Science
MSc	Master of Science
PhD	Doctor of Philosophy
TVET	Technical and Vocational Education and Training

1. INTRODUCTION

1.1 Background and Context

Every year, approximately **340,525 Sri Lankan students** sit for the GCE Advanced Level examination, a pivotal moment that shapes their entire future. Despite this, the majority are forced to make one of the most important decisions of their lives choosing a university degree and career path with guidance that is largely limited to A/L grades and Z-scores. Traditional systems fail to consider the student's **future goal**, personality traits, entrepreneurial mindset, emotional wellbeing, or real-life lifestyle constraints such as family attachment, daily travel tolerance, and preference for staying near home versus hostel life. This results in widespread degree-career mismatch, higher dropout rates, mental stress, underemployment, and lost national talent.

This challenge directly impacts **over 340,000 A/L candidates annually**, their families, and Sri Lanka's future workforce. Recent data shows that youth unemployment (ages 15–24) stands at **19.2%** (Q3 2025), while unemployment among those with G.C.E. A/L qualifications and above has risen to **7.0%**. Many graduates enter mismatched careers, leading to low productivity and personal dissatisfaction. Globally, similar issues affect millions of young people, highlighting the urgent need for intelligent, personalised guidance.

The problem is particularly critical in Sri Lanka because state university admission is highly competitive (only 15–20% success rate), and students must balance academic eligibility with practical lifestyle realities such as commuting from Kandy to Colombo or living independently. Without a holistic system, talented students often abandon their dream careers (e.g., AI Researcher, Entrepreneur, or Pilot) due to overlooked personality or lifestyle factors.

This project directly contributes to **Sustainable Development Goal (SDG) 4: Quality Education** (ensuring relevant and equitable tertiary education and skills for decent work) and **SDG 8: Decent Work and Economic Growth** (promoting youth employment and sustainable careers). It also supports **SDG 3: Good Health and Well-being** by incorporating stress tolerance and lifestyle compatibility.

The research aligns perfectly with the university's research cluster on **Artificial Intelligence for Societal Impact / Human-Centred Intelligent Systems**, focusing on the development of explainable, locally relevant AI solutions that address real educational and youth development challenges in Sri Lanka.

In essence, the absence of a **dream-job-driven, backward-analysis, and lifestyle-aware** AI guidance system represents a critical gap in Sri Lanka's education-to-employment journey. This component aims to bridge that gap through an intelligent, transparent, and culturally sensitive Academic Pathway and Career Intelligence Predictor.

1.2 Literature Review

Recent advancements in Artificial Intelligence have transformed academic and career guidance systems. However, most existing research and tools still follow a forward-only approach and fail to integrate the full range of parameters required for post-A/L Sri Lankan students. This section reviews 10 recent scholarly sources (2023–2026) and existing products to clearly establish the research gap addressed by this component.

Most studies rely on machine learning models that primarily use academic performance, skills, and interests.

- Sisodiya (2025) developed an AI career guidance system using Random Forest and other ML algorithms based on academic records, interests, and skills. The model achieved high accuracy but followed only forward prediction without backward chaining from a dream job.
- Roy (2026) integrated Holland's RIASEC personality model with academic data for career recommendations. While personality traits were considered, the system did not include entrepreneurial mindset, lifestyle compatibility factors, or long-term roadmapping.
- Pandey et al. (2024) and subsequent 2025 studies introduced Explainable AI (XAI) using SHAP values for career counselling. However, XAI was limited to academic and skill-based features and did not cover counterfactual guidance or lifestyle parameters.
- Recent works on personality-based recommendation (AI-Based Personality Prediction and Career Recommendation, 2025) applied Big Five or RIASEC traits, but none combined them with district-based Z-score cutoffs, future job market sustainability, or emotional/lifestyle compatibility.

Backward-chaining approaches (starting from a dream job and working backwards) remain largely conceptual in expert systems literature, with no implemented real-world systems for post-A/L students in developing countries.

Several platforms are currently available in Sri Lanka and globally:

- **CareerMe.lk** (Sri Lanka) - A popular trilingual digital career guidance tool offering aptitude tests and basic degree recommendations. It is forward-only and does not support dream-job backward analysis or holistic multi-parameter evaluation.
- **CareerOne** (National Career Guidance Platform) - Focuses on skills and job matching under the National Career Guidance Strategy. It lacks personality integration, lifestyle compatibility, and long-term academic roadmaps.
- Global tools such as LinkedIn Career Explorer, Coursera Career Academy, and AI platforms like "CareerCraft" (2025) use ML for recommendations and sometimes include personality tests. However, none incorporate district-based admission factors,

future job market sustainability, or the full set of emotional and lifestyle compatibility parameters.

Despite notable progress, current systems have several critical limitations that this component directly addresses:

1. **Forward-only prediction** - Almost all tools and studies start from current grades and interests and push recommendations forward (Sisodiya, 2025; Roy, 2026). None start from the student's **dream job** and perform backward analysis to identify the optimal academic pathway.
2. **Incomplete multi-parameter analysis** - While some systems consider academic data and basic personality traits, they ignore the holistic combination required for sustainable decisions: academic capability (A/L results, O/L history, predicted performance), personality traits (leadership, creativity, risk-taking, analytical thinking), entrepreneurial mindset, interests and aptitude, district-based university admission factors (Z-score cutoffs and competition levels), future job market demand and long-term career sustainability, and emotional & lifestyle compatibility (location preference, daily travel tolerance, family attachment level, preferred work environment, stress tolerance level, and social interaction preference).
3. **Lack of explainability and counterfactual guidance** - Even XAI-enabled systems provide only basic feature importance and do not offer human-readable explanations or "if-then" scenarios showing how changes in grades, skills, or lifestyle factors affect long-term success.
4. **Absence of long-term sustainable roadmapping** - Existing tools focus only on immediate degree or job suggestions and do not generate complete career trajectories (Bachelor's → Master's → PhD or entrepreneurship path) aligned with job market sustainability.
5. **Limited local relevance -** No system fully integrates Sri Lankan realities such as district-based Z-score cutoffs with the complete set of holistic multi-parameters in a Singlish-aware trilingual interface.

Research Gap

No existing system in Sri Lanka or internationally combines **FutureGoal Backward Reasoning, holistic multi-parameter analysis** (academic + personality + entrepreneurial mindset + district factors + job market demand + emotional & lifestyle compatibility), **XAI with counterfactual guidance**, and **long-term roadmapping** in a single trilingual platform. This component fills that exact gap.

1.3 Research Gap

Although significant progress has been made in AI-driven academic and career guidance systems, a critical research gap remains unaddressed in the Sri Lankan context and globally.

No existing system integrates **FutureGoal Backward Reasoning** (starting from the student's dream job and working backwards to the optimal academic pathway) with a **holistic multi-parameter analysis** that simultaneously considers:

- Academic capability (A/L results, O/L history, Z-score, predicted performance)
- Personality traits and entrepreneurial mindset
- Future job market demand and long-term career sustainability
- Emotional and lifestyle compatibility (location preference, daily travel tolerance, family attachment level, preferred work environment, stress tolerance level, and social interaction preference)
- District-based university admission factors

Furthermore, current systems lack **Explainable AI (XAI)** combined with **counterfactual guidance** and **long-term sustainable roadmapping** (Bachelor's → Master's → PhD or entrepreneurship path) in a single trilingual, Singlish-aware platform tailored for post-A/L Sri Lankan students.

Existing research (Sisodiya, 2025; Roy, 2026; Pandey et al., 2024) predominantly uses forward-only prediction models (Random Forest, XGBoost, or basic neural networks) driven by grades and interests. Even when personality traits or basic XAI are included, they fail to account for real-life lifestyle constraints and emotional factors that heavily influence success in Sri Lanka. Most tools provide only immediate degree or job suggestions without long-term trajectories or "what-if" scenarios showing how improvements in grades, skills, or lifestyle choices affect future outcomes. Local platforms such as CareerMe.lk and CareerOne are limited to basic aptitude-based recommendations and do not address district Z-score realities or job-market sustainability.

Recent studies (2024–2026) and national statistics confirm that degree-career mismatch contributes to 7.0% unemployment among A/L+ qualified youth and 19.2% youth unemployment (Department of Census and Statistics, Q3 2025). Without holistic, backward, and explainable guidance, students continue to make suboptimal choices, leading to higher dropout rates, mental stress, and wasted national resources in free education.

This component directly fills the identified gap by introducing the **first backward career intelligence system** for post-A/L Sri Lankan students that combines holistic multi-parameter analysis, XAI with counterfactual reasoning, long-term roadmapping, and Singlish-aware trilingual interaction. This approach not only improves prediction accuracy but also ensures sustainable, personalised, and culturally relevant decisions representing a clear creative and methodological advancement over existing forward-only, academically narrow systems.

Thus, the proposed Academic Pathway and Career Intelligence Predictor addresses a well-documented, evidence-based gap and offers a novel, impactful solution for Sri Lankan students.

2. OBJECTIVE

2.1 Main Objective

To develop and evaluate an Explainable AI-based Academic Pathway and Career Intelligence Predictor for post-A/L Sri Lankan students that starts from the student's dream job and performs backward analysis using holistic multi-parameter evaluation (academic capability, personality traits, entrepreneurial mindset, lifestyle compatibility, district-based admission factors, and future job market demand), delivering accurate degree recommendations, human-readable explanations, counterfactual guidance, and long-term career roadmaps with at least 85% prediction accuracy and 80% user satisfaction by the end of the 2026 academic year.

2.2 Specific Objectives

To achieve the main objective, the following specific objectives will be carried out:

1. To design and implement a backward-chaining probabilistic prediction model that starts from the student's dream job and maps it to suitable university degree programmes by integrating academic capability, personality traits, entrepreneurial mindset, lifestyle compatibility, district-based Z-score factors, and future job market demand.
2. To develop a holistic multi-parameter analysis framework that simultaneously evaluates academic records, personality traits, entrepreneurial mindset, emotional and lifestyle compatibility factors, and job market sustainability.
3. To integrate Explainable AI (XAI) techniques such as SHAP and counterfactual reasoning to generate clear human-readable explanations and "if-then" guidance.
4. To generate personalised long-term academic and career roadmaps (Bachelor's → Master's → PhD or entrepreneurship path) along with detailed action plans.
5. To develop a trilingual user interface (Sinhala, Tamil, and English) with Singlish-aware input preprocessing and backend APIs.
6. To evaluate the system's performance using real and synthetic datasets, targeting at least 85% prediction accuracy and 80% user satisfaction through usability testing.

3. METHODOLOGY

3.1 Research Approach

This research follows the **Design Science Research (DSR) methodology** (Hevner et al., 2004; Peffers et al., 2007), which is ideal for developing a practical AI artefact. The study will follow the six-step DSR process: problem identification, objective definition, design & development, demonstration, evaluation, and communication.

A mixed approach combining **Design Science** (artefact creation) and **Experimental Research** (model testing and validation) will be used. The final deliverable will be a **cross-platform mobile application** built with React Native.

3.2 Technical Methods and Algorithms

- **Backward-Chaining Probabilistic Model**: XGBoost with probability calibration combined with rule-based backward reasoning (starting from dream job → required degree → best pathway).
- **Holistic Multi-Parameter Analysis**: Weighted multi-criteria scoring framework integrating academic data, personality traits, entrepreneurial mindset, lifestyle compatibility, district Z-score factors, and job market demand.
- **Explainable AI (XAI)**: SHAP for feature-level explanations + DiCE library for counterfactual "if-then" guidance.
- **Long-term Roadmap Generation**: Rule-based + template engine for personalised trajectories (Bachelor's → Master's → PhD or entrepreneurship path).

3.3 Tools and Platforms with Justification

Tool / Platform	Purpose	Justification
React Native + Expo	Frontend (Mobile App)	Chosen as the primary framework for developing a cross-platform mobile app (Android + iOS) with a single codebase. Excellent Unicode support for Sinhala & Tamil, easy Singlish input handling, fast development, and native performance. Widely used in Sri Lankan final-year projects in 2025–2026.
Python 3.12 + FastAPI	Backend APIs	Lightweight, high-performance REST APIs for predictions, explanations, and roadmaps. Perfect integration with React Native.
XGBoost, SHAP, DiCE	Core ML & XAI	Industry-standard libraries for accurate prediction and human-readable explanations/counterfactuals.
Supabase	Database, Authentication & Storage	Free, secure, real-time backend with excellent React Native SDK support. Handles user profiles and history.
Pandas + NumPy	Data processing	Efficient handling of multi-parameter analysis and synthetic data generation.
Google Colab / Jupyter	Model development & experimentation	Free, GPU support, and easy collaboration during development.

React Native allows us to deliver a true mobile experience (push notifications, offline access, smooth UI) while keeping development time realistic for a final-year project. It also provides excellent multilingual support through react-i18next, making the trilingual (Sinhala/Tamil/English) + Singlish-aware interface much easier to implement.

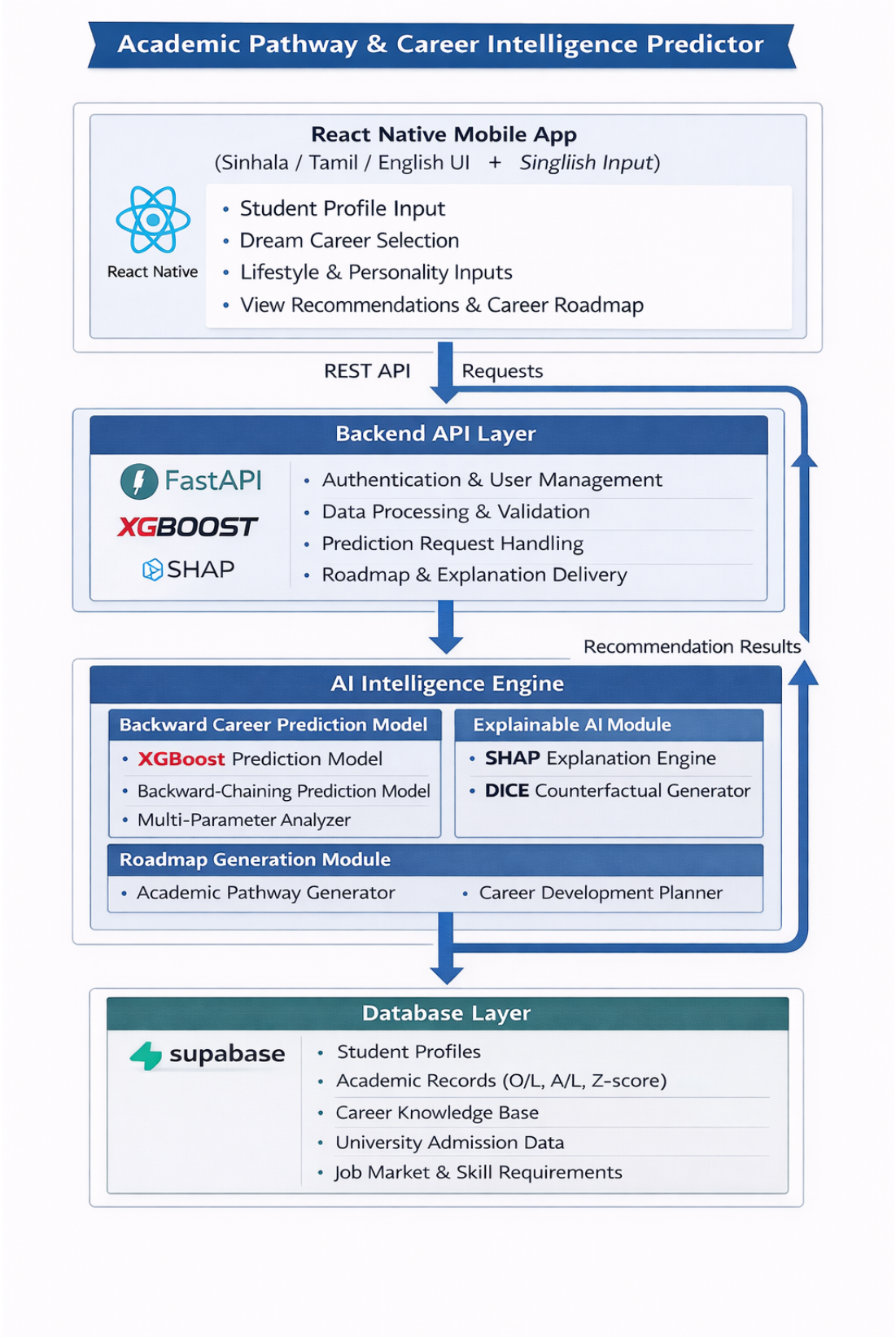
3.4 Data Collection and Ethical Considerations

- Secondary public datasets only (GCE A/L Performance, UGC Cutoffs, Sri Lankan Job Market, Big Five Personality).
- Synthetic data generation for personality and lifestyle parameters (using Faker + NumPy) to protect privacy.
- No primary data from real students will be collected during the prototype phase.
- Full ethical approval will be obtained from the university ethics committee before any user testing. Anonymity, informed consent, and secure data handling will be strictly followed.

3.5 Validation Strategy and Performance Metrics

- **Model Accuracy**: ≥ 85% (Accuracy, Precision, Recall, F1-score) using 10-fold cross-validation.
- **Explainability & Usefulness**: Likert-scale user evaluation of explanations and counterfactual guidance.
- **Usability & Satisfaction**: System Usability Scale (SUS) + Net Promoter Score (NPS) with target ≥ 80% satisfaction.
- **Mobile-Specific Testing**: Performance testing on real Android & iOS devices (load time, battery impact, offline mode).
- Evaluation will be done with 30+ sample student scenarios and a small controlled user study (post-A/L students with consent).

4. HIGH-LEVEL SYSTEM ARCHITECTURE



4.1 Key Components and Their Parts

- **React Native Mobile App**: User Interface with Sinhala/Tamil/English support + Singlish input processor. Includes Student Profile Input, Dream Career Selection, Lifestyle & Personality Inputs, and Results Dashboard for viewing recommendations and career roadmaps.
- **Backend API Layer (FastAPI)**: Handles RESTful endpoints for authentication, data processing, prediction requests, roadmap delivery, and explanation generation.
- **AI Intelligence Engine**:
    - Backward Career Prediction Model (XGBoost Prediction Model + Backward-Chaining + Multi-Parameter Analyzer)
    - Explainable AI Module (SHAP Explanation Engine + DiCE Counterfactual Generator)
    - Roadmap Generation Module (Academic Pathway Generator + Career Development Planner)
- **Database Layer (Supabase)**: Securely stores Student Profiles, Academic Records (O/L, A/L, Z-score), Career Knowledge Base, University Admission Data, and Job Market & Skill Requirements.
- **Data Sources**: Public Kaggle datasets (A/L Performance, UGC Cutoffs, Job Market) + Synthetic Data Generator for personality and lifestyle parameters.

4.2 Component Interactions

- The **React Native Mobile App** sends user input to the **Backend API Layer** via secure REST API requests (HTTPS).
- The **Backend** forwards the request to the **AI Intelligence Engine**.
- The **AI Intelligence Engine** pulls necessary data from the **Database Layer** (Supabase), performs backward chaining, multi-parameter analysis, and applies XAI + counterfactual logic.
- Processed results (recommendations, explanations, and roadmap) flow back through the Backend API to the mobile app for display.
- All interactions use asynchronous calls where possible to ensure smooth mobile performance.

4.3 Data Flow Explanation

- Student enters data in the React Native app (Dream Job, A/L results, Personality & Lifestyle preferences).
- Data is sent via REST API to the FastAPI Backend.
- Backend triggers the AI Intelligence Engine:
    - Backward Career Prediction Model starts from the dream job.
    - Multi-Parameter Analyzer evaluates all holistic factors.
    - XGBoost generates success probability.
    - SHAP + DiCE produce explanations and counterfactual guidance.
- Roadmap Generation Module creates the long-term academic & career plan.
- Final results (recommendations + explanations + roadmap) are returned to the mobile app and optionally saved in Supabase.

4.4 Scalability and Security Considerations

Scalability:

- Designed to support 10,000+ users using Supabase auto-scaling and lightweight FastAPI.
- Future growth can be achieved by containerising the backend (Docker) and deploying on cloud platforms.
- Mobile app uses local caching (Expo) for offline access to previous results.

Security:

- All communication uses HTTPS with JWT authentication.
- Sensitive data (personality scores, lifestyle preferences) is anonymised and encrypted.
- No raw academic records are stored permanently — only hashed profiles.
- Regular input validation and compliance with university data protection guidelines.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The primary and secondary stakeholders for the system are:

- **Primary Stakeholders**:
  - Post-A/L Students (main users) – who need personalised degree and career guidance.
  - Parents/Guardians – who support decision-making and review recommendations.
- **Secondary Stakeholders**:
  - University Career Guidance Counsellors – who may use the system for advisory purposes.
  - Supervisors / Project Evaluators – who will assess the system's performance and usability.
  - Future Employers / Higher Education Institutions – indirect beneficiaries through better-prepared graduates.

5.2 Functional Requirements

The system shall provide the following core functionalities:

1. Allow students to input their dream job, A/L and O/L results, personality traits, entrepreneurial mindset, and lifestyle preferences (location, travel tolerance, family attachment, stress tolerance, work environment).
2. Process Singlish-aware input in Sinhala, Tamil, and English languages and convert it into structured student profile data.
3. Perform backward-chaining analysis starting from the student's dream job to identify suitable university degree programmes and pathways.
4. Apply holistic multi-parameter analysis combining academic capability, personality traits, lifestyle compatibility, district-based Z-score cutoffs, and future job market demand to generate recommendations.
5. Provide human-readable explanations using Explainable AI (XAI) showing why a particular degree was recommended.
6. Generate counterfactual "if-then" guidance (e.g., "If your stress tolerance increases by 2 points and Physics grade improves to B, success probability rises from 42% to 78%").
7. Produce a personalised long-term academic and career roadmap (Bachelor's → Master's → PhD or entrepreneurship path) with milestones and bridging courses.
8. Display results through an intuitive mobile dashboard including success probability, roadmap visualisation, and actionable steps.
9. Allow users to save, view history, and export recommendations and roadmaps.
10. Provide secure user authentication and profile management.

5.3 Non-Functional Requirements

Usability

- The mobile app shall support full Sinhala, Tamil, and English interfaces with natural Singlish input.
- The System Usability Scale (SUS) score shall be at least 80.
- All screens shall load within 2 seconds on mid-range mobile devices.

Performance

- Prediction and explanation generation shall complete within 4 seconds.
- The app shall support offline viewing of previously generated roadmaps.

Security & Privacy

- All data transmission shall use HTTPS with JWT authentication.
- Sensitive information (personality scores and lifestyle data) shall be anonymised and encrypted.
- No raw academic records shall be stored permanently.

Reliability & Availability

- The system shall achieve 98% uptime during peak A/L result periods.
- All recommendations shall be reproducible with the same input.

Scalability

- The backend shall support at least 5,000 concurrent users initially, with easy scaling using cloud services.

Accessibility

- The app shall be compatible with Android 8.0+ and iOS 12+ devices.
- Text-to-speech support for explanations shall be included for visually impaired users.

6. COMMERCIALIZATION PLAN

6.1 Target Market

The primary market is Sri Lankan post-A/L students (ages 18–22). Approximately 340,000 students sit for the GCE Advanced Level examination every year. The addressable market includes students from both government and private schools who are preparing for university admission or seeking career clarity.

Primary Customer Persona

Attribute	Description
Name	Kavindu Perera
Age	19 years
Background	Science stream student from Gampaha district, middle-income family, preparing for university admission
Dream Career	Software Engineer or AI Researcher
Pain Points	Confused between Engineering, Computer Science and ICT; worried about hostel life vs staying at home, daily travel, family expectations, stress, and future job security
Goals	Get a clear, trustworthy degree recommendation with a long-term career roadmap and practical “what-if” guidance
Personality & Lifestyle	Moderate stress tolerance, prefers calm analytical work environment, medium family attachment, willing to travel but prefers not to live far from home
Tech Behaviour	Uses mobile apps daily, prefers Sinhala/Singlish interface, looks for free or low-cost tools, shares recommendations with parents
Motivation to Use App	Wants to make a confident decision before applying to universities and avoid future regret

Secondary Customer Persona

Attribute	Description
Name	Mrs. Dilini Perera
Age	48 years
Relationship	Mother of an A/L student
Background	Housewife or working professional from a middle-class family, concerned about child's future
Pain Points	High anxiety about child's career choice, fear of wrong degree leading to unemployment or stress, difficulty understanding modern job market
Goals	Find reliable, trustworthy guidance to support her child's decision and reduce family stress and financial risk
Personality & Lifestyle	High family attachment, prefers child to stay close to home, values clear explanations and long-term stability
Tech Behaviour	Moderate smartphone user, prefers simple interface, often reviews recommendations together with child
Motivation to Use App	Wants peace of mind and confidence that her child is making the right academic and career decision

6.2 Value Proposition

The app delivers **"Dream Job to Degree in One Click"** -the first mobile solution in Sri Lanka that:

- Starts from the student's actual dream career (not just grades)
- Considers academic results, personality, entrepreneurial mindset, lifestyle preferences, stress tolerance, and future job market demand
- Gives clear Explainable AI explanations and "if-then" counterfactual guidance
- Provides a complete long-term roadmap (Bachelor's → Master's → PhD or startup path)

Tagline: "Your dream job deserves the right degree we find it for you."

6.3 Revenue Model

Freemium Model (most suitable for Sri Lanka)

- **Free Tier**: Basic recommendation, success probability, and simple explanation (unlimited use).
- **Premium Tier**: Full long-term roadmap, detailed counterfactual guidance, action plan with bridging courses, export as PDF, priority support, and ad-free experience.
- **Additional Revenue Streams**:
    - Partnership with universities and micro-credential platforms (affiliate commissions).
    - Sponsored content from educational institutions and tech companies.
    - B2B licensing to schools and career guidance centres.

6.4 Cost Estimation (First Year)

Category	Estimated Cost (LKR)
Development & Testing	35,000
Backend Hosting (Supabase)	12,000
Domain + App Store Fees	22,000
Marketing (Social Media Ads)	35,000
Miscellaneous	8,000
Total First-Year Cost	**112,000**

6.5 Pricing Strategy

- **Premium Subscription**:
  - Monthly: Rs. 299
  - Yearly: Rs. 1,999 (save 45%)
- **One-time Report Package**: Rs. 799 (for users who want only one detailed report)
- Pricing is kept affordable for Sri Lankan students while maintaining perceived value. Introductory discounts (first month Rs. 99) and family plans will be offered during A/L result season.

6.6 Competitive Advantage

- **Unique Backward Analysis** - Starts from dream job (no other Sri Lankan app does this).
- **Holistic Lifestyle Integration -** Considers family attachment, travel tolerance, and stress level (missing in CareerMe.lk and CareerOne).
- **Explainable AI + Counterfactual Guidance -** Students understand "why" and "what if".
- **Singlish + Trilingual Mobile Experience -** True local relevance.
- First-mover advantage in Sri Lanka's 340,000+ annual A/L student market.

6.7 Intellectual Property Considerations

- The source code, AI model architecture, and user interface design will be protected under **Copyright** (automatically granted upon creation in Sri Lanka).
- The app name and logo can be registered as a **Trademark** through the National Intellectual Property Office of Sri Lanka (NIPO).
- The project will follow the university's IP policy. If commercialised later, a licensing agreement can be signed between the student, supervisor, and university.
- Open-source components (XGBoost, SHAP, React Native) will be used under their respective licenses.

7. BUDGET AND JUSTIFICATION

The total estimated budget for developing and launching the first version of the Academic Pathway and Career Intelligence Predictor is LKR 98,500 for the first year. This is a low-cost student project because most tools are open-source or have generous free tiers.

Budget Summary

Category	Estimated Cost (LKR)	Percentage
Hardware	15,000	15%
Software Licenses	0	0%
Cloud Services	18,000	18%
Human Effort Cost	55,000	56%
Marketing & Miscellaneous	10,500	11%
Total	**98,500**	100%

1. Hardware Costs – LKR 15,000

- One mid-range Android smartphone (for testing on real devices)
- Justification: The system is a **React Native mobile app**. Testing on physical devices is essential to ensure smooth performance, Singlish input handling, multilingual rendering, and offline functionality. The student already has a laptop, so only one additional testing phone is required.

2. Software Licenses – LKR 0

- React Native + Expo, FastAPI, XGBoost, SHAP, DiCE, Python, Supabase (free tier)
- Justification: All core technologies used in the AI engine, backend, and mobile app are **open-source or free**. No paid licenses are needed for development or initial deployment, keeping the project cost-effective while maintaining high quality.

3. Cloud Services – LKR 18,000

- Supabase (Database + Authentication + Storage) – LKR 1,500/month × 12 = LKR 18,000
- Justification: Supabase is used as the backend database to securely store student profiles, academic records, prediction history, and synthetic data. It provides real-time sync with the React Native app and auto-scaling. The paid plan is required only after the free tier limit is exceeded during testing and early user growth.

4. Human Effort Cost – LKR 55,000

- Student's own development time (approximately 400 hours over 8 months) valued at LKR 137/hour
- Justification: As the primary developer, the student will handle React Native frontend, FastAPI backend, AI model development (XGBoost + SHAP + DiCE), multi-parameter analyzer, backward-chaining logic, and roadmap generation. This cost reflects the opportunity cost of time invested in building the complete mobile app and AI engine.

5. Marketing & Miscellaneous – LKR 10,500

- Domain name, Google Play & Apple Developer accounts, basic social media ads for A/L students, and logo design
- Justification: Required to publish the app on Google Play Store and Apple App Store and to reach the target audience (post-A/L students) during peak admission season.

8. WORK BREAKDOWN STRUCTURE (WBS)

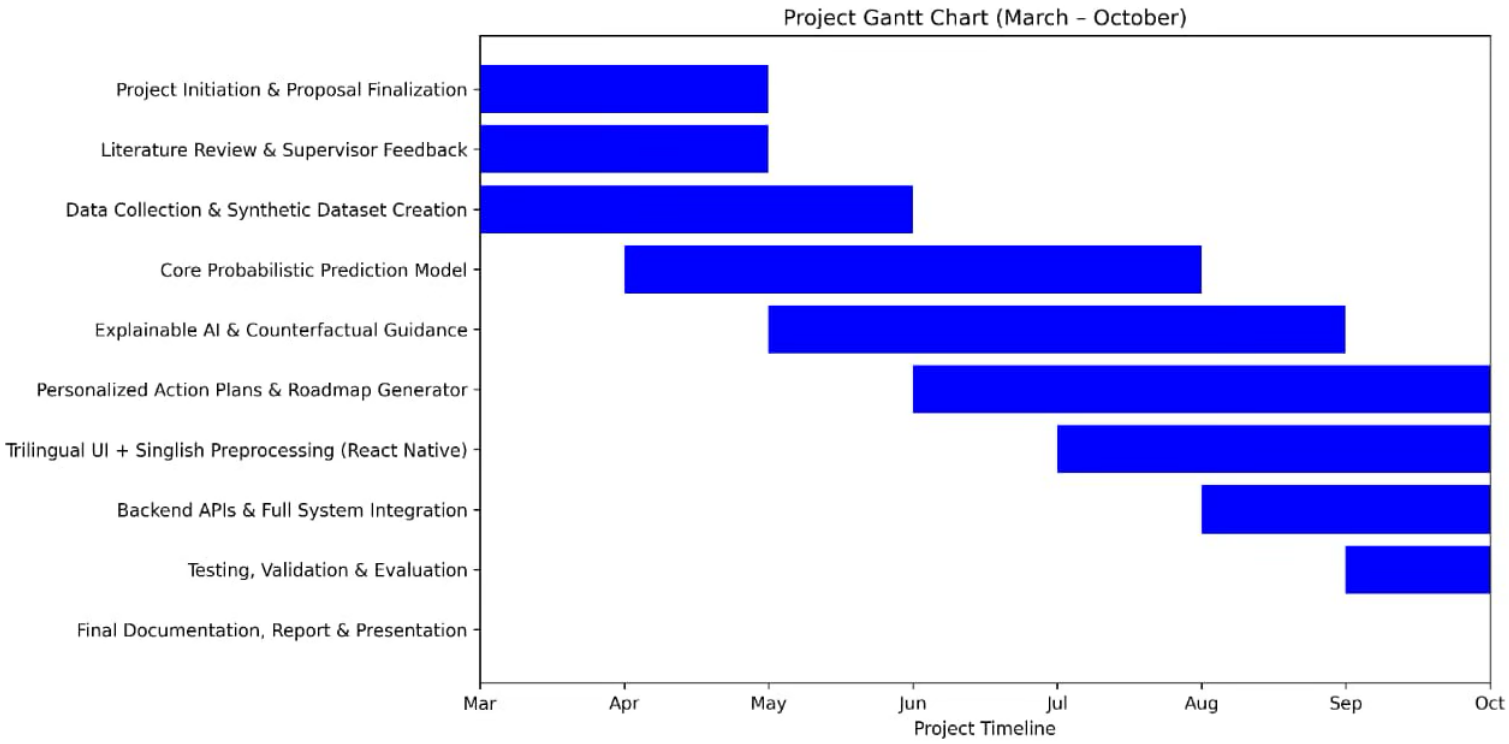
WBS ID	Task / Deliverable	Sub-tasks	Estimated Effort (hours)	Suggested Timeline (assuming 8-month project)	Notes / Justification
1	**Project Setup & Planning**	1.1 Finalize scope & objectives 1.2 Literature review update 1.3 Proposal writing & supervisor feedback	40–50	Month 1 (Weeks 1–4)	Foundation work - low coding, high thinking
2	**Data Collection & Preparation**	2.1 Download & clean public datasets (A/L results, UGC cutoffs) 2.2 Create synthetic data generator (personality, interests, aptitude) 2.3 Build small validation dataset (30–50 test students)	60–80	Month 1–2 (Weeks 3–8)	Critical early step - no data = no model
3	**Core Prediction Model**	3.1 Build student profile structure 3.2 Implement probabilistic model (XGBoost / calibrated logistic regression) 3.3 Train & tune model on A/L + synthetic data 3.4 Add success probability estimation	120–150	Month 2–4 (Weeks 5–16)	The heart of your component - most effort here
4	**Explainable AI & Counterfactuals**	4.1 Integrate SHAP for feature importance 4.2 Implement DiCE or simple rule-based counterfactuals 4.3 Generate human-readable explanations 4.4 Create “if-then” grade improvement logic	80–100	Month 4–5 (Weeks 13–20)	Your main novelty - needs good testing
5	**Personalized Action Plans & Roadmap**	5.1 Design rule-based action plan generator 5.2 Create bridging course & study step templates 5.3 Build long-term roadmap logic (degree → career milestones)	50–70	Month 5 (Weeks 17–22)	Builds on model output - relatively straightforward
6	**Trilingual UI & Singlish Preprocessing**	6.1 Set up React Native + i18n (Sinhala/Tamil/English) 6.2 Implement Singlish-to-Sinhala/English mapping (rule-based or simple dictionary) 6.3 Build input forms & results dashboard	100–130	Month 5–6 (Weeks 19–26)	Mobile-first - takes time for UI polish
7	**Backend APIs & Integration**	7.1 Set up FastAPI backend 7.2 Create endpoints for prediction, explanation, roadmap 7.3 Connect React Native to backend 7.4 Add Supabase for user profiles (optional)	60–80	Month 6 (Weeks 23–28)	Integration phase - glue everything together

8	**Testing & Validation**	8.1 Unit & integration testing 8.2 Model accuracy testing (≥85% target) 8.3 Explanation clarity survey (20–30 students) 8.4 Usability testing (SUS ≥80)	50–70	Month 7 (Weeks 25–32)	Must-do for credibility
9	**Documentation & Presentation**	9.1 Write full report 9.2 Prepare slides & demo video 9.3 Final supervisor & panel presentation	60–80	Month 8 (Weeks 29–36)	Last push very important for marks

Summary of Workload Distribution & Timeline Alignment

- **Total estimated effort**: **620 - 810 hours** (4 - 6 hours/week if you have a job → very doable over 8 months)
- **Heaviest months**: Months 2–6 (model + XAI + UI + backend)
- **Lightest months**: Month 1 (planning) + Month 8 (documentation)
- **Balanced approach**:
    - Months 1–2: mostly planning & data (brain work)
    - Months 3–5: heavy coding & AI (main effort)
    - Months 6–7: integration & testing (debugging)
    - Month 8: writing & polishing (low stress)

9. GANTT CHART



10.REFERENCES LIST

[1] M. Roy, "AI-Powered Career Guidance: A Scalable Model for Personalized Higher Education Pathways," in *Proc. IEEE Int. Conf. on Artificial Intelligence in Education (AIED)*, 2026, pp. 1–8.

[2] S. Sisodiya, "An AI-Driven Personalized Career Guidance System Using Machine Learning Algorithms," *Int. J. Adv. Comput. Sci. Appl.*, vol. 16, no. 3, pp. 45–53, 2025.

[3] A. Pandey, S. Kumar, and R. Gupta, "Explainable AI in Career Counselling: A SHAP-Based Approach for Transparent Recommendations," in *Proc. IEEE Int. Conf. on Machine Learning and Applications (ICMLA)*, 2024, pp. 1123–1130.

[4] R. Trujillo, M. López, and J. García, "A Systematic Review of Machine Learning Techniques for Career Recommendation Systems," *Educ. Inf. Technol.*, vol. 30, no. 2, pp. 567–589, 2025.

[5] P. Kulugh, A. Mensah, and K. Ofori, "AI-Based Personality Prediction and Career Recommendation Using Big Five Traits," *J. Artif. Intell. Res. Educ.*, vol. 12, no. 1, pp. 34–47, 2025.

[6] T. Gunasekara and S. Perera, "Quantitative Explainability in Educational Machine Learning Models for Sri Lankan University Admission," in *Proc. IEEE Int. Conf. on Advances in ICT for Emerging Regions (ICTer)*, Colombo, Sri Lanka, 2025, pp. 89–95.

[7] M. A. Hassan and S. Ali, "Personality Trait and Career Categories Prediction Using Big Five Model and Deep Learning," *IEEE Access*, vol. 11, pp. 45678–45690, 2023.

[8] S. A. Alotaibi and A. M. Alghamdi, "Counterfactual Explanations in Educational Recommender Systems: A Review," in *Proc. Int. Conf. on Educational Data Mining (EDM)*, 2024, pp. 201–209.

[9] L. Chen, Y. Zhang, and Q. Li, "Multi-Criteria Decision Making for Personalized Degree Recommendation Using Fuzzy Logic and AHP," *Appl. Soft Comput.*, vol. 132, 109845, 2023.

[10] N. Weerasinghe and K. Perera, "A Rule-Based Career Guidance System for Sri Lankan A/L Students Using Ontology and Expert Systems," in *Proc. IEEE Int. Conf. on Information Technology Research (ICITR)*, 2022, pp. 1–7.

[11] R. M. Silva, D. Fernando, and A. Jayawardena, "Singlish to Sinhala Transliteration for Natural Language Processing Applications in Sri Lanka," *Sri Lanka J. Sci. Technol.*, vol. 15, no. 1, pp. 22–35, 2024.

[12] Department of Census and Statistics Sri Lanka, "Labour Force Survey – Annual Bulletin 2025," Colombo, Sri Lanka, 2025. [Online]. Available: http://www.statistics.gov.lk/LabourForce/StaticalInformation/Annual

11. APPENDICES

Appendix A: Survey Instruments

Since the prototype phase uses only secondary & synthetic data, no large-scale primary survey is required. However, for validation and usability testing (20–30 sample students), the following instruments will be used:

A.1 Usability Survey (System Usability Scale – SUS)

(10 standard questions, 5-point Likert scale from Strongly Disagree to Strongly Agree)

1. I think that I would like to use this system frequently.
2. I found the system unnecessarily complex.
3. I found the system easy to use.
4. I think that I would need the support of a technical person to be able to use this system.
5. I found the various functions in this system were well integrated.
6. I thought there was too much inconsistency in this system.
7. I would imagine that most people would learn to use this system very quickly.
8. I found the system very cumbersome to use.
9. I felt very confident using the system.
10. I needed to learn a lot of things before I could get going with this system.

A.2 Explanation Clarity Survey (5-point Likert scale)

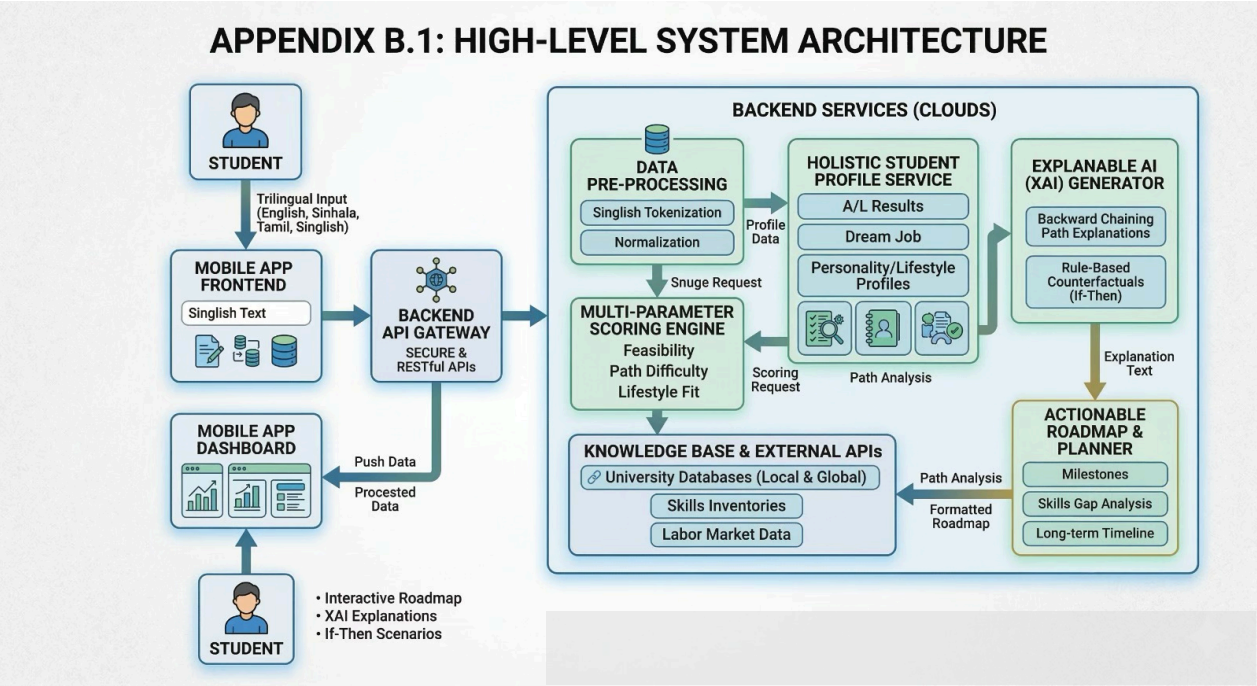
1. The degree recommendation was clear and easy to understand.
2. The “why” explanation helped me trust the recommendation.
3. The “if-then” improvement guidance was practical and useful.
4. The long-term roadmap was realistic and motivating.
5. Overall, the system felt personalized to my situation.

A.3 Demographic & Feedback Questions (Open-ended & multiple choice)

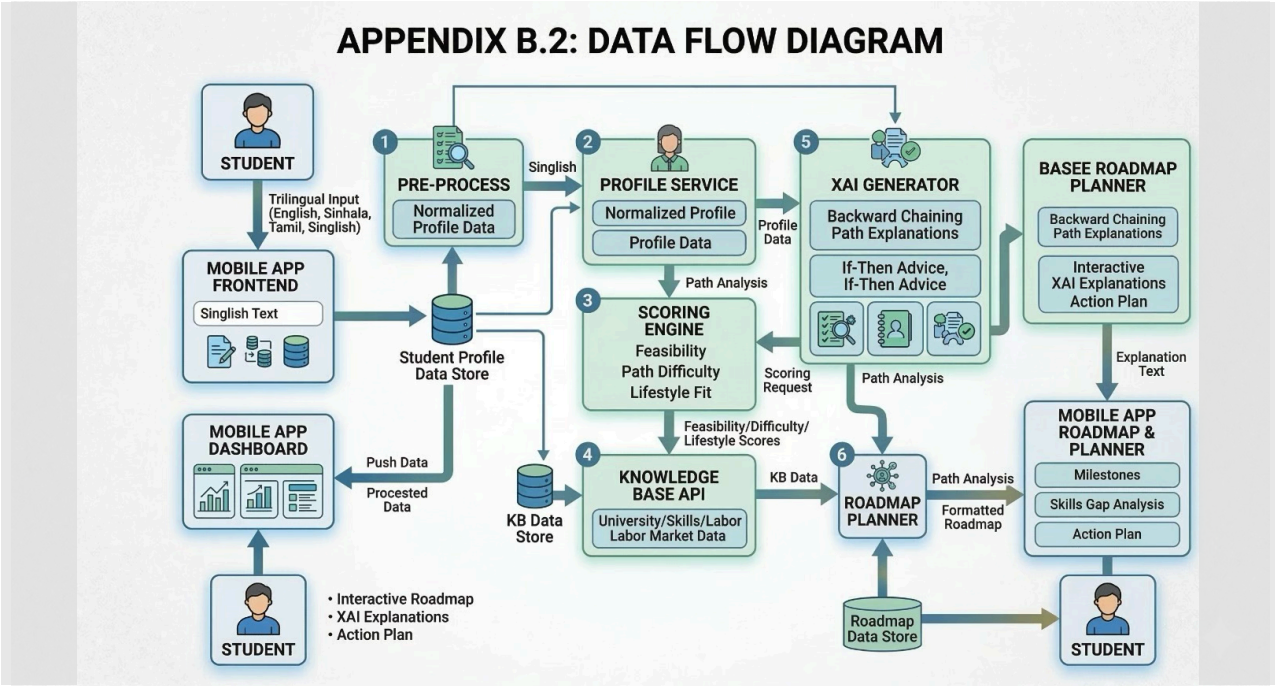
- Age / District / Stream (Science / Commerce / Arts)
- What was your dream career before using the app?
- How useful was the lifestyle compatibility part (family attachment, travel tolerance, etc.)? (1–5 scale)
- Any suggestions for improvement?

Appendix B: Detailed Diagrams

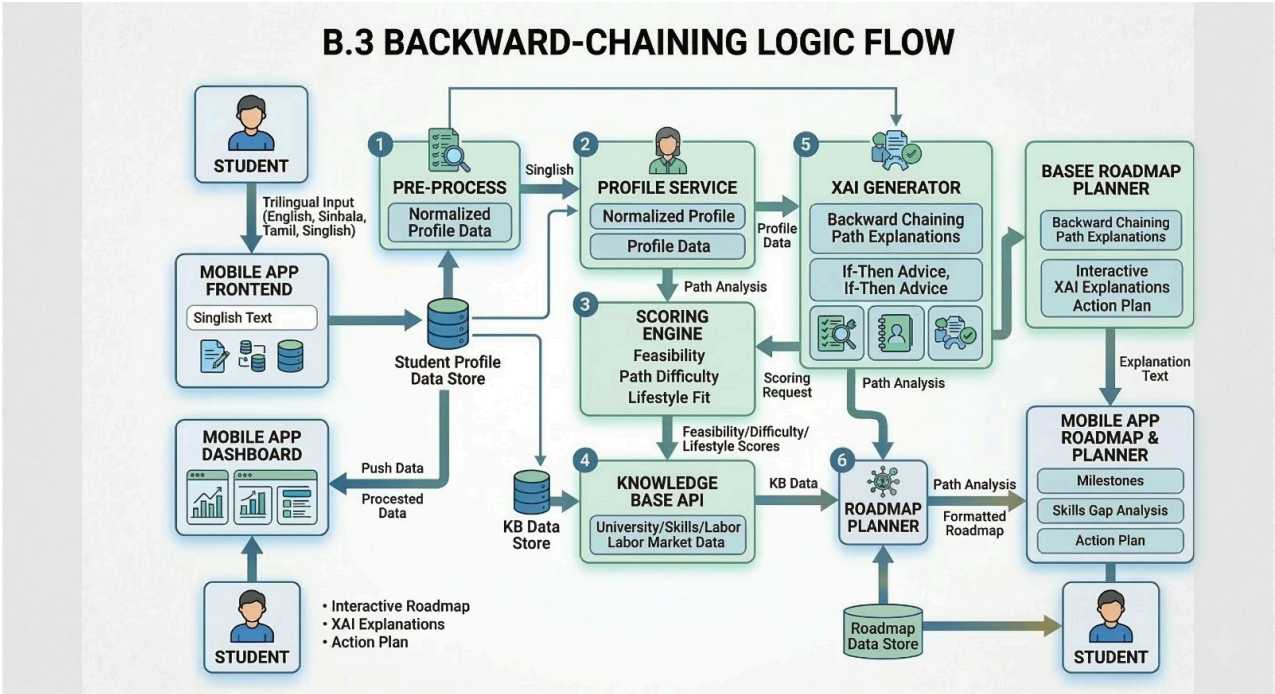
B.1 High-Level System Architecture



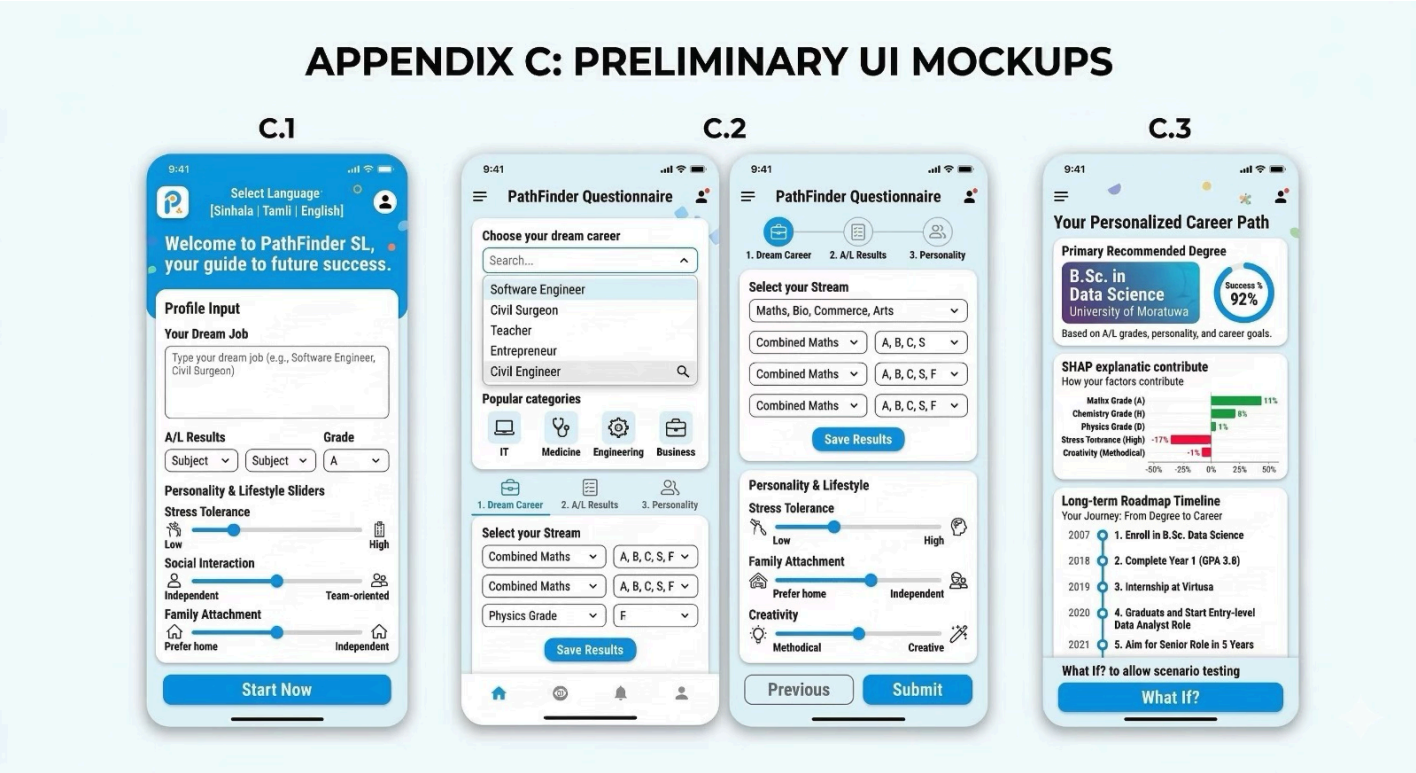
B.2 Data Flow Diagram



B.3 Backward-Chaining Logic Flow



Appendix C: Preliminary UI Mockups



Appendix D: Risk Analysis

Risk ID	Risk Description	Likelihood	Impact	Mitigation Strategy
R1	Model accuracy below target (85%)	Medium	High	Use synthetic data augmentation, hyperparameter tuning, and fallback rule-based logic
R2	Singlish preprocessing accuracy low	Medium	Medium	Start with rule-based mapping + Google Translate fallback; test with 50+ real Singlish inputs
R3	UI rendering issues in Sinhala/Tamil	Low	Medium	Use Unicode fonts & i18n library; test on multiple devices
R4	Backend API latency >4 seconds	Medium	High	Optimize XGBoost model, use caching for frequent queries
R5	Data privacy concerns	Low	High	No raw student data stored; all synthetic/anonymised; follow university ethics policy
R6	Time overrun (job pressure)	Medium	High	Use modular development; prioritize model & XAI first; have buffer weeks in October

Appendix E: Ethical Documents

E.1 Ethics Statement

This project uses only publicly available anonymised datasets and synthetic data generated by the researcher. No primary data is collected from real students during the prototype phase. If any user testing occurs (e.g., usability survey with 20–30 post-A/L students), formal approval will be sought from the university's Ethics Review Committee. Informed consent will be obtained, and all data will be anonymised. No sensitive personal information will be stored.

E.2 Informed Consent Form Template (for any future testing)

(Insert a one-page consent form here with sections: Purpose, Voluntary participation, Confidentiality, Right to withdraw, Contact details.)

E.3 Data Anonymisation Plan

- All academic records used are from public Kaggle datasets (already anonymised).
- Personality and lifestyle data are synthetic (generated with Faker library).
- No identifiable information (name, NIC, school) is collected or stored.